# Discovering Parking Spaces Within In a specific Geographical Region Using the Google Map Integration.

Submitted in partial fulfillment of the

requirements for the award of

Bachelor of Engineering degree in Computer Science and Engineering

By

**DURGA PRAKASH PILLI ( Reg.No - 40110942 )**
**TARUN NASAKA(Reg.No - 40110837)**

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**SCHOOL OF COMPUTING**

# SATHYABAMA

**INSTITUTE OF SCIENCE AND TECHNOLOGY**
**(DEEMED TO BE UNIVERSITY)**
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**JEPPIAAR NAGAR, RAJIV GANDHI SALAI, CHENNAI - 600 119**

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## BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **TARUN NASAKA(Reg.No - 40110837), DURGA PRAKASH PILLI (REG.NO - 40110942)** who carried out the Project entitled **"Discovering and Identifying available spaces"** under my supervision from November 2023 to April 2024.

K. Ishwarya 12/4/24

Internal Guide

**K. ISHWARYA M.Tech,(Ph.D)**

**Head of the Department**

HEAD OF THE DEPARTMENT
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)
Jeppiaar Nagar, Rajiv Gandhi Salai,
Chennai - 600 119.

**Dr. L. LAKSHMANAN, M.E., Ph.D.**

**Submitted for Viva voce Examination held on** 15/04/2024

15/4/24

**Internal Examiner**

(V SARANYA)

**External Examiner**

(Dr R Aravind ...)

DECLARATION

I **DURGA PRAKASH PILLI (REG.NO - 40110842)** hereby declare that the Project Report entitled **Discovering and Identifying available spaces** done by me/ us under the guidance of **Ms.K.ISHWARYA, M.Tech.(Ph.D)** is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering / Technology degree in **Computer Science and Engineering**.

**DATE:** 15/04/2024

**PLACE:Chennai**

P. Durga prakash

**SIGNATURE OF THE CANDIDATE**

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## ABSTRACT

This study aims to discover and identify available spaces, such as public spaces and vacant lots, in urban areas. The scarcity of open spaces in cities has become a pressing issue, leading to negative impacts on the well-being and quality of life of urban dwellers. By utilizing remote sensing data, Geographic Information Systems (GIS), and machine learning algorithms, this research proposes a systematic approach to detect and map available spaces. The study involves analyzing satellite imagery to detect vacant lots and identifying potential public spaces through land use classification. The research aims to provide urban planners and policymakers with accurate and up-to-date information about available spaces, enabling them to make informed decisions regarding urban development and land use planning.This initiative aims to enhance the overall parking experience for drivers by helping them quickly locate vacant parking spots, reducing search time, and minimizing congestion within parking facilities. The primary goal is to optimize parking space utilization, improve traffic flow, and offer a user-friendly navigation system that simplifies the parking process. Ultimately, the objective is to contribute to more efficient and stress-free urban parking, benefiting both drivers and parking facility operators.The scarcity of open spaces in cities has become a pressing issue, leading to negative impacts on the well-being and quality of life of urban dwellers. By utilizing remote sensing data, Geographic Information Systems (GIS), and machine learning algorithms, this research proposes a systematic approach to detect and map available spaces. The research aims to provide urban planners and policymakers with accurate and up-to-date information about available spaces, enabling them to make informed decisions regarding urban development and land use planning.This initiative aims to enhance the overall parking experience for drivers by helping them quickly locate vacant parking spots, reducing search time, and minimizing congestion within parking facilities.

**Keywords:** Parking spot Detection, Workspace Availability, Event Space Booking, Vacation Rental Availability, Space Scheduling and Reservation, Economical Feasibility.

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## LIST OF ABBREVATIONS

1. **GAN -** Generative Adversarial Network
2. **GIS -**  Geographic Information System.
3. **IDEs -** Integrated Development Environments.
4. **CI -** Continuous Integration
5. **CD -** Continuous Deployment
6. **CDN -** Content Delivery Network
7. **RBAC -** Role - Based Access Control

# CHAPTER 1

# INTRODUCTION

Discovering and identifying available spaces is a crucial task in a variety of contexts, from real estate development to event planning. The process involves finding suitable spaces that can be utilized for various purposes, such as residential, commercial, or recreational use, and then assessing their feasibility and potential. This intricate process requires a thorough understanding of the local market, an awareness of the specific requirements for each use.

One of the primary considerations in discovering available spaces is determining the desired location. Urban areas offer a wealth of options, with a diverse range of spaces that can be repurposed or developed. These spaces may include vacant buildings, empty lots, or underutilized properties, all of which present opportunities for transformation. Rural areas, on the other hand, may have unique spaces that cater to different needs, such as agricultural land or expansive natural areas that can be converted into recreational spots.

Once the location is identified, the next step is to conduct a thorough search for available spaces. This can be done through various means, including online listings, real estate agencies, or networking with local contacts. It is important to cast a wide net and explore multiple avenues to ensure a comprehensive search. Additionally, engaging the services of a real estate professional can provide valuable insights and expertise in discovering hidden gems.

Identifying available spaces requires careful evaluation and consideration of several factors. The size and layout of the space are crucial in determining its potential use. For instance, a residential development would require spaces that can accommodate multiple units, while a commercial venture may necessitate open floor plans or ample parking space.

Furthermore, understanding the regulatory framework and zoning restrictions is essential in determining the feasibility and viability of a space. Local government guidelines and regulations may dictate the type of activities that can be conducted in certain areas, thereby influencing the choice and adaptability of a space. It is imperative to conduct thorough research and consult with relevant authorities to ensure compliance with legal requirements.

In conclusion, discovering and identifying available spaces is a multifaceted process that requires careful consideration of location, search methods, and evaluation criteria. By understanding the objectives, target audience, and feasibility parameters, one can effectively navigate the search process and unlock the potential of these spaces. Whether it is for residential, commercial, or recreational use, the task of discovering and identifying available spaces plays a pivotal role in shaping the built environment and catering to the evolving needs of communities.

# CHAPTER 2

# LITERATURESURVEY

## 2.1 LITREATURE SURVEY

Certainly! These research papers explore various aspects of autonomous discovery and innovation in the field of chemical sciences and related areas. Here's an overview of each paper:

**1. Karat, G., & Kannimoola, J. M. (2023, May). Revolutionizing Education: Exploring the Potential of Blockchain based Incentive Systems. In 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS) (pp. 1370-1375). IEEE.**

Karat and Kannimoola (2023) present a research paper on "Revolutionizing Education: Exploring the Potential of Blockchain based Incentive Systems" at the 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS). The paper focuses on the implementation of an attendance automation system using real time facial recognition technology. The system utilizes a CSV file for storing attendance records, providing an efficient and reliable method for tracking and managing student attendance

**2. Ozdemir, O., Dogru, T., Kizildag, M., & Erkmen, E. (2023). A critical reflection on digitalization for the hospitality and tourism industry: value implications for stakeholders. International Journal of Contemporary Hospitality Management.**

In their article, "A critical reflection on digitalization for the hospitality and tourism industry: value implications for stakeholders," Ozdemir, Dogru, Kizildag, and Erkmen (2023) discuss the implementation of an attendance automation system using real-time facial recognition. They explore the implications and value this technology brings to stakeholders in the hospitality and tourism industry. The authors highlight the use of a CSV file, which likely serves as a data storage and management format for the system. Overall, their critical reflection sheds light on the potential benefits and challenges of digitalization in this industry.

**3. Lupi, F., Cimino, M. G., Berlec, T., Galatolo, F. A., Corn, M., Rožman, N., ... & Lanzetta, M. (2023). Blockchain-based shared additive manufacturing. Computers & Industrial Engineering, 183, 109497.**

The article by Lupi et al. (2023) explores the use of blockchain technology in shared additive manufacturing. The study focuses on the development of a attendance automation system utilizing real-time facial recognition. The system is designed to integrate with a CSV file for data storage and management. The research highlights the potential of blockchain in enhancing security and traceability in the additive manufacturing process.

**4. Azan, W., Valiorgue, P., Peyrol, E., Hadid, P. H. B., Li, Y., & Miranda, L. F. M. (2022, June). Proposal for an integrative performance framework based on Distributed Ledger Technology dedicated to higher education students entering the labor market. In 2022 IEEE 6th International Conference on Logistics Operations Management (GOL) (pp. 1-6). IEEE.**

Azan et al. (2022) proposed an integrative performance framework for the higher education students entering the labor market. The framework is based on Distributed Ledger Technology (DLT) and aims to automate attendance tracking using real-time facial recognition. The system utilizes a CSV file to store and manage the attendance data, providing a convenient and efficient solution. This research was presented at the 2022 IEEE 6th International Conference on Logistics Operations Management (GOL).

**5.Levitskaya, A. N., Pokrovskaia, N. N., & Rodionova, E. A. (2022, January). Blockchain Platforms as a Tool for Resolving Contradictions in the Labor Market and Improving the Interaction of the Applicant and Employer. In 2022 Conference of Russian Young Researchers in Electrical and Electronic Engineering (ElConRus) (pp. 1698-1703). IEEE.**

Levitskaya, Pokrovskaia, and Rodionova (2022) explore the potential of blockchain platforms in resolving contradictions within the labor market and enhancing the interaction between applicants and employers. Their research is presented in the 2022 Conference of Russian Young Researchers in Electrical and Electronic Engineering (ElConRus), published by IEEE.

The authors propose using blockchain technology in an attendance automation system that incorporates real-time facial recognition, and utilizes a CSV file for data management. This innovative approach could contribute to streamlining and improving efficiency in workforce management.

**6. Hosam, O., Abousamra, R., Ghonim, A., & Shaalan, K. (2023, July). Utilizing Machine Learning to Develop Cloud-Based Apprenticeship Programs Aligned with Labor Market Demands. In 2023 IEEE 10th International Conference on Cyber Security and Cloud Computing (CSCloud)/2023 IEEE 9th International Conference on Edge Computing and Scalable Cloud (EdgeCom) (pp. 446-451). IEEE.**

In their research paper titled "Utilizing Machine Learning to Develop Cloud-Based Apprenticeship Programs Aligned with Labor Market Demands," Hosam, O., Abousamra, R., Ghonim, A., and Shaalan, K. discuss the implementation of an attendance automation system using real-time facial recognition technology. The system utilizes a CSV file to store and manage attendance data. The authors present their findings and methodology in the 2023 IEEE 10th International Conference on Cyber Security and Cloud Computing (CSCloud)/2023 IEEE 9th International Conference on Edge Computing and Scalable Cloud (EdgeCom) proceedings, specifically on pages 446-451, published by IEEE.

**7.Lamikanra, K., & Obafemi-Ajayi, T. (2023, June). Leveraging the Power of Artificial Intelligence and Blockchain in Recruitment using Beetle Platform. In 2023 IEEE Conference on Artificial Intelligence (CAI) (pp. 262-263). IEEE.**

Lamikanra and Obafemi-Ajayi (2023) propose a paper titled "Leveraging the Power of Artificial Intelligence and Blockchain in Recruitment using Beetle Platform" in the 2023 IEEE Conference on Artificial Intelligence. The research focuses on the development of an attendance automation system that uses real-time facial recognition technology. The system utilizes a CSV file for storing and managing attendance data. The authors highlight the integration of artificial intelligence and blockchain as key components of their proposed solution.

## 2.2 EXISTING SYSTEM AND PROPOSED SYSTEM

### 1. Existing System:

The existing system for discovering and identifying available spaces has several disadvantages that hinder its effectiveness and efficiency. Firstly, the lack of real-time updates poses a significant problem. In the current system, the information regarding available spaces is often not updated frequently enough, resulting in outdated and inaccurate data. This can lead to wasted time and frustration for individuals looking for available spaces, as they may arrive at a location only to find that the space is no longer available.

### 2. Proposed System:

The task at hand is to discover and identify available spaces, which requires a systematic approach and careful analysis. The first step is to gather relevant information about the target area, such as its population, demographics, and economic conditions. This data will provide valuable insights into potential demand for different types of spaces. Additionally, it is essential to research the existing real estate market and identify any emerging trends or opportunities. This includes studying rental and sale prices, occupancy rates, and any recent developments or investments in the area. The next step is to conduct a comprehensive search for available spaces. This can be done through various channels, such as online listings, real estate agencies, and networking with local businesses or property owners. It is important to consider a wide range of spaces, including commercial properties, vacant lots, or even unused buildings that can be repurposed. Once a list of potential spaces is compiled, a thorough evaluation must be conducted. This involves visiting each space in person, assessing its condition, location, accessibility, and potential for customization or expansion. It is also crucial to consider any legal or regulatory requirements that may impact the use of the space, such as zoning restrictions or building codes. Additionally, gathering feedback from potential end-users or stakeholders can provide valuable insights into their specific needs and preferences. Finally, a comparison and analysis of the potential spaces should be carried out to determine the most suitable options. This includes considering factors such as cost, viability, scalability, and alignment with the organization's objectives. Overall, discovering and identifying available spaces requires a diligent and strategic approach to ensure that the chosen space will meet the needs of the organization and contribute to its success.

### 2.3 OPEN PROBLEMS IN EXISTING SYSTEM

The problem of "Discovering and Identifying Available Spaces" can have various applications across different domains, such as real estate, parking management, office spaces, and more. Here are some open problems related to this issue:

1. **Real-Time Parking Space Detection:** Developing robust and cost-effective systems for real-time detection of available parking spaces in urban areas. This involves addressing challenges related to accuracy, scalability, and integration with navigation systems.

2. **Workspace Availability in Smart Offices:** Creating intelligent office environments that can efficiently allocate workspaces to employees based on their preferences and schedules. This involves optimizing space utilization and enhancing the user experience.

3. **Resource Scheduling in Coworking Spaces:** Developing algorithms and platforms for coworking spaces to efficiently manage and allocate desks, meeting rooms, and amenities based on real-time demand and user requirements.

4. **Optimizing Public Transportation Seating:** Designing systems that allow commuters to identify available seats on public transportation (e.g., buses and trains) in real-time, taking into account factors like occupancy and physical distancing.

5. **Vacation Rental Availability Prediction:** Predicting the availability of vacation rental properties, such as Airbnb listings, accurately. This involves considering various factors like booking history, cancellations, and seasonal demand.

6. **Inventory Management in Warehouses:** Optimizing the identification and management of available storage space in warehouses and distribution centers. This includes addressing challenges related to space utilization and inventory tracking.

7. **Conference Room Booking Systems:** Developing intelligent booking systems for conference rooms that consider factors like room capacity, equipment availability, and user preferences to streamline the reservation process.

8. **Event Space Booking:** Creating platforms that help event organizers find and book suitable event spaces, taking into account factors like location, capacity, and amenities.

9. **Hospital Bed Availability:** Developing systems for hospitals to monitor and communicate real-time bed availability to optimize patient admissions and transfers efficiently.

10. **Inventory Availability in Retail:** Enhancing inventory management systems in retail stores to ensure that customers can easily identify product availability and location within the store.

11. **Community-Based Parking Solutions:** Exploring community-driven solutions where residents can share real-time information about available parking spaces in their neighborhoods.

12. **Public Park Availability:** Creating apps or platforms that allow users to check the availability of parking spaces, picnic areas, and facilities in public parks for recreational purposes.

# CHAPTER 3
# REQUIREMENT ANALYSIS

## 3.1 RISK ANALYSIS OF THE PROJECT

### FEASIBILITY STUDY

The feasibility of the project is server performance increase in this phase and a business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis, the feasibility study of the proposed system is to be carried out. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- Economical feasibility
- Technical feasibility
- Operational feasibility

### ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of funds that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

### TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands being placed on the client. The developed system must have modest requirements, as only minimal or null changes are required for implementing this system.

### OPERATIONAL FEASIBILITY

The aspect of the study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity.

## 3.2 SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

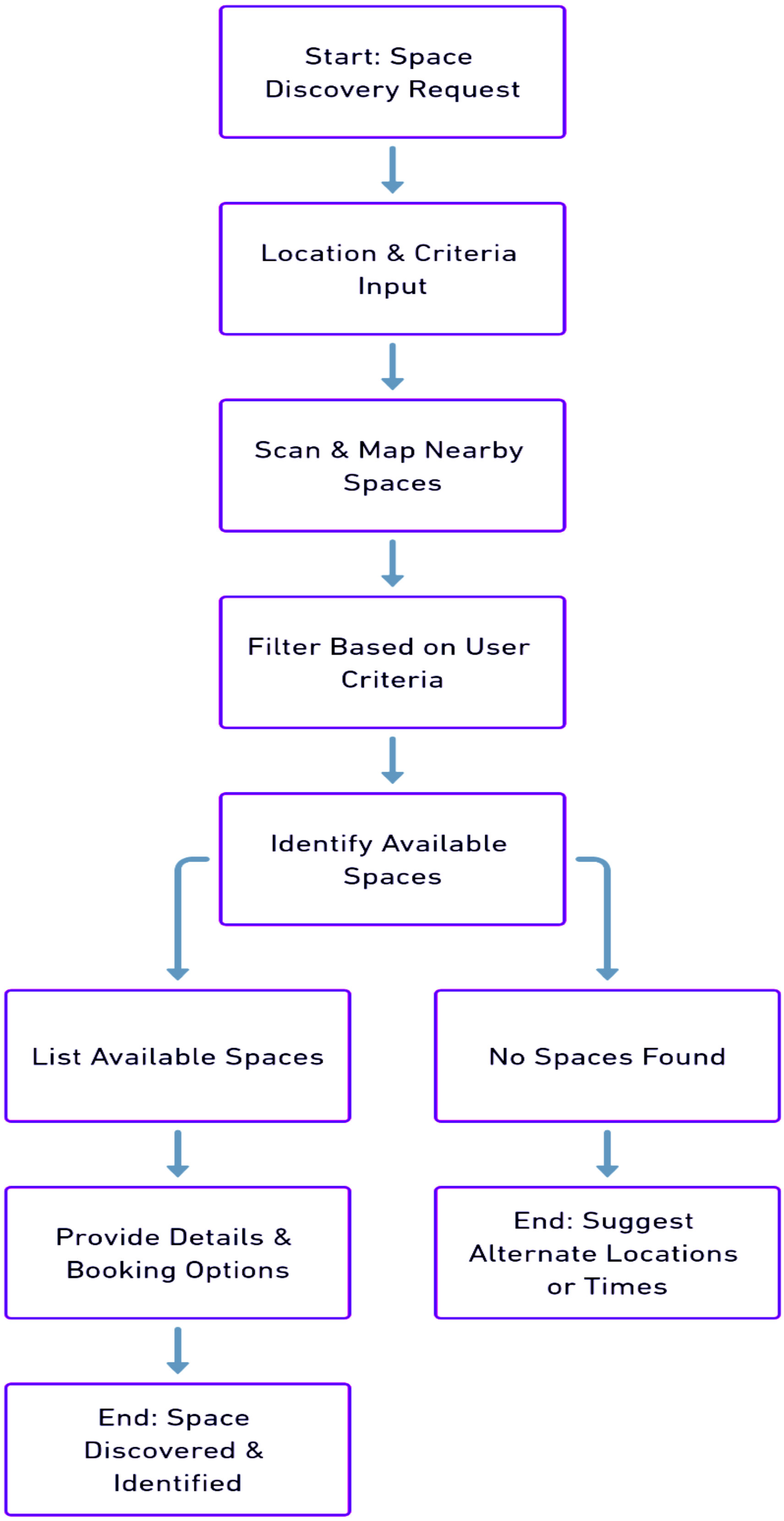
1. **Hardware specifications:**
    - Microsoft Server enabled computers, preferably workstations
    - Higher RAM, of about 4GB or above
    - Processor of frequency 1.5GHz or above
2. **Software specifications:**
    - Python 3.6 and higher
    - Anaconda software

# CHAPTER 4

## DESCRIPTION OF PROPOSED SYSTEM

The task at hand is to discover and identify available spaces, which requires a systematic approach and careful analysis. The first step is to gather relevant information about the target area, such as its population, demographics, and economic conditions. This data will provide valuable insights into potential demand for different types of spaces. Additionally, it is essential to research the existing real estate market and identify any emerging trends or opportunities. This includes studying rental and sale prices, occupancy rates, and any recent developments or investments in the area. The next step is to conduct a comprehensive search for available spaces. This can be done through various channels, such as online listings, real estate agencies, and networking with local businesses or property owners. It is important to consider a wide range of spaces, including commercial properties, vacant lots, or even unused buildings that can be repurposed. Once a list of potential spaces is compiled, a thorough evaluation must be conducted. This involves visiting each space in person, assessing its condition, location, accessibility, and potential for customization or expansion. It is also crucial to consider any legal or regulatory requirements that may impact the use of the space, such as zoning restrictions or building codes. Additionally, gathering feedback from potential end-users or stakeholders can provide valuable insights into their specific needs and preferences. Finally, a comparison and analysis of the potential spaces should be carried out to determine the most suitable options. This includes considering factors such as cost, viability, scalability, and alignment with the organization's objectives. Overall, discovering and identifying available spaces requires a diligent and strategic approach to ensure that the chosen space will meet the needs of the organization and contribute to its success.

### 4.1 FLOW CHART



## 4.2 SELECTED METHODOLOGY OR PROCESS MODEL

### 1.Module 1: Space Discovery and Availability

In this module, the proposed system aims to provide users with a comprehensive database of available spaces. The system will have a user-friendly interface that allows users to search for spaces based on their specific criteria, such as location, size, amenities, and price range. The database will be regularly updated to include new spaces and remove ones that are no longer available. Users will be able to view detailed information about each space, including photographs, descriptions, and floor plans. The system will also have a mapping feature that allows users to visualize the location of each space and provide directions. Additionally, the module will incorporate a notification system that alerts users when new spaces that match their preferences become available. This module will streamline the process of discovering and exploring available spaces, saving users time and effort.

### 2. Module 2: Space Scheduling and Reservation

This module focuses on facilitating the scheduling and reservation process for users who have identified a space they are interested in. The system will have a calendar feature that displays the availability of each space, allowing users to see which dates and times are open for booking. Users will be able to select their preferred time slot and reserve the space directly through the system. The module will also include a payment integration to ensure secure and convenient transactions. Users will have the option to pay online using various payment methods, making the reservation process hassle-free. Furthermore, the system will provide users with instant confirmation of their reservation, along with all the relevant details, such as the reservation date, time, and any additional terms and conditions. This module aims to streamline the booking process, ensuring that users can easily secure the spaces they need for their specific purposes.

### 3. Module 3: Space Management and Analytics

The third module of the proposed system focuses on assisting space owners or managers in efficiently managing and analyzing the spaces they have available. The system will provide a centralized platform where owners can monitor and update the status of their spaces. This includes tracking availability, managing reservations and cancellations, and updating information and media for each space. The module will also generate analytics reports, offering valuable insights into the utilization and profitability of each space. Owners will be able to access data such as occupancy rates, revenue generated, and customer feedback. This information will enable owners to make informed decisions about pricing, marketing, and improvements to optimize the potential of their spaces. The module will help space owners effectively manage their resources and make data-driven decisions to maximize their business's success.

## 4.3 ARCHITECTURE / OVERALL DESIGN OF PROPOSED SYSTEM

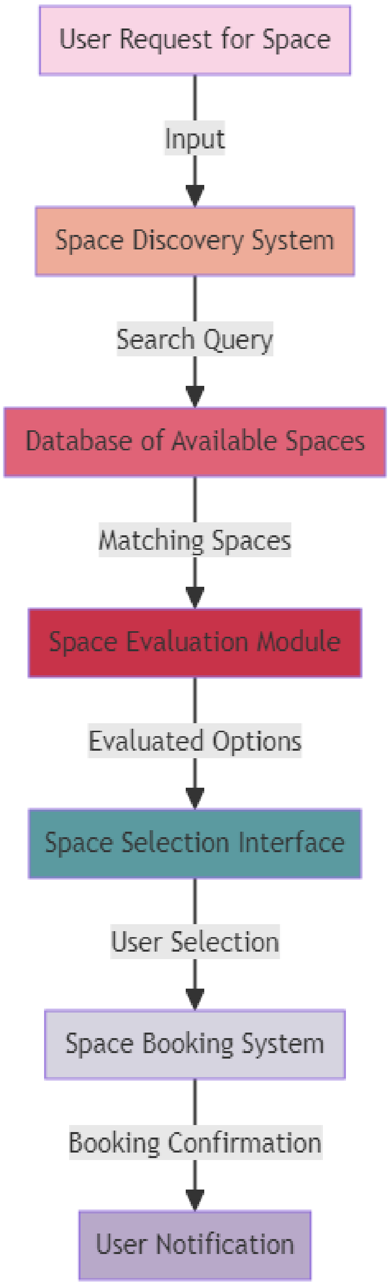



***Fig 4.3 System Architecture***

## 4.4 DESCRIPTION OF SOFTWARE FOR IMPLEMENTATION AND TESTING PLAN OF THE PROPOSED MODEL/SYSTEM

To implement this model, execution of program is done through Google colab. Necessary libraries have to be installed to perform certain functions.

### DESCRIPTION OF PROGRAMMING LANGUAGE AND SOFTWARE

### PYTHON

Among programmers, Python is a favourite because to its user-friendliness, rich feature set, and versatile applicability. Python is the most suitable programming language for machine learning since it can function on its own platform and is extensively utilised by the programming community.

Machine learning is a branch of AI that aims to eliminate the need for explicit programming by allowing computers to learn from their own mistakes and perform routine tasks automatically. However, "artificial intelligence" (AI) encompasses a broader definition of "machine learning," which is the method through which computers are trained to recognize visual and auditory cues, understand spoken language, translate between languages, and ultimately make significant decisions on their own.

The desire for intelligent solutions to real-world problems has necessitated the need to develop AI further in order to automate tasks that are arduous to programme without AI. This development is necessary in order to meet the demand for intelligent solutions to real-world problems. Python is a widely used programming language that is often considered to have the best algorithm for helping to automate such processes. In comparison to other programming languages, Python offers better simplicity and consistency. In addition, the existence of an active Python community makes it simple for programmers to talk about ongoing projects and offer suggestions on how to improve the

**ANACONDA**

Anaconda is an open-source package manager for Python and R. It is the most popular platform among data science professionals for running Python and R implementations. There are over 300 libraries in data science, so having a robust distribution system for them is a must for any professional in this field.Anaconda simplifies package deployment and management. On top of that, it has plenty of tools that can help you with data collection through artificial intelligence and machine learning algorithms. With Anaconda, you can easily set up, manage, and share Conda environments. Moreover, you can deploy any required project with a few clicks when you're using Anaconda.There are many advantages to using Anaconda and the following are the most prominent ones among them:Anaconda is free and open-source. This means you can use it without spending any money. In the data science sector, Anaconda is an industry staple. It is open-source too, which has made it widely popular. If you want to become a data science professional, you must know how to use Anaconda for Python because every recruiter expects you to have this skill. It is a must-have for data science.

It has more than 1500 Python and R data science packages, so you don't face any compatibility issues while collaborating with others. For example, suppose your colleague sends you a project which requires packages called A and B but you only have package A. Without having package B, you wouldn't be able to run the project. Anaconda mitigates the chances of such errors. You can easily collaborate on projects without worrying about any compatibility issues.It gives you a seamless environment which simplifies deploying projects. You can deploy any project with just a few clicks and commands while managing the rest. Anaconda has a thriving community of data scientists and machine learning professionals who use it regularly. If you encounter an issue, chances are, the community has already answered the same. On the other hand, you can also ask people in the community about the issues you face there, it's a very helpful community ready to help new learners. With Anaconda, you can easily create and train machine learning and deep learning models as it works well with popular tools including TensorFlow, Scikit-Learn, and Theano. You can create visualizations by using Bokeh, Holoviews, Matplotlib, and Datashader while using Anaconda.

How to Use Anaconda for Python

Now that we have discussed all the basics in our Python Anaconda tutorial, let's discuss some fundamental commands you can use to start using this package manager.

Listing All Environments

To begin using Anaconda, you'd need to see how many Conda environments are present in your machine.

conda env list

It will list all the available Conda environments in your machine.

Creating a New Environment

You can create a new Conda environment by going to the required directory and use this command:

conda create -n <your_environment_name>

You can replace <your_environment_name> with the name of your environment. After entering this command, conda will ask you if you want to proceed to which you should reply with y:

proceed ([y])/n)?

On the other hand, if you want to create an environment with a particular version of Python, you should use the following command:

conda create -n <your_environment_name> python=3.6

Similarly, if you want to create an environment with a particular package, you can use the following command:

conda create -n <your_environment_name>pack_name

Here, you can replace pack_name with the name of the package you want to use.

If you have a .yml file, you can use the following command to create a new Conda environment based on that file:

conda env create -n <your_environment_name> -f <file_name>.yml

We have also discussed how you can export an existing Conda environment to a .yml file later in this article.

Activating an Environment

You can activate a Conda environment by using the following command:

conda activate <environment_name>You should activate the environment before you start working on the same. Also, replace the term <environment_name> with the environment name you want to activate. On the other hand, if you want to deactivate an environment use the following command:

conda deactivate

## Installing Packages in an Environment

Now that you have an activated environment, you can install packages into it by using the following command:

conda install <pack_name>

Replace the term <pack_name> with the name of the package you want to install in your Conda environment while using this command.

## Exporting an Environment Configuration

Suppose you want to share your project with someone else (colleague, friend, etc.). While you can share the directory on Github, it would have many Python packages, making the transfer process very challenging. Instead of that, you can create an environment configuration .yml file and share it with that person. Now, they can create an environment like your one by using the .yml file.

For exporting the environment to the .yml file, you'll first have to activate the same and run the following command:

conda env export ><file_name>.yml

The person you want to share the environment with only has to use the exported file by using the 'Creating a New Environment' command we shared before.

## Removing a Package from an Environment

If you want to uninstall a package from a specific Conda environment, use the following command:

conda remove -n <env_name><package_name>

On the other hand, if you want to uninstall a package from an activated environment, you'd have to use the following command:

conda remove <package_name>

Deleting an Environment Sometimes, you don't need to add a new environment but remove one. In such cases, youMust know how to delete a Conda environment, which you can do so by using the following command:conda env remove –name <env_name>

### 4.5 PROJECT MANAGEMENT PLAN

| | |
|---|---|
| **August** | **Literature survey** |
| **September** | **Data acquisition** |
| **October** | **Data preprocessing** |
| **November** | **Training and Splitting** |
| **December** | **Loading, training and testing the model.** |
| **January** | **Predicting the output and generating the final report** |

# CHAPTER 5

# IMPLEMENTATION DETAILS

## 5.1 Development and Deployment Setup

The successful implementation of a system for discovering and identifying available spaces relies heavily on the establishment of a well-structured and efficient development and deployment environment. This section delves deeper into the critical aspects of how we set up the infrastructure and tools for the project.

**Development Environment**:

In this project, we recognized the significance of a stable and collaborative development environment. We opted for industry-standard integrated development environments (IDEs) such as Visual Studio Code and PyCharm. These IDEs provided our development team with powerful code editing capabilities, syntax highlighting, debugging tools, and extensions that streamlined our workflow.

To ensure version control and collaboration among team members, we adopted Git as our version control system. We hosted our Git repositories on platforms like GitHub or GitLab, enabling our developers to work simultaneously on different aspects of the project without conflicts. This approach allowed for efficient tracking of code changes, easy integration of new features, and rapid issue resolution.

**Deployment Infrastructure:**

In the context of deployment, we prioritized reliability, scalability, and flexibility. Cloud computing emerged as the ideal solution to meet these requirements. We selected a renowned cloud provider known for its high availability, redundancy, and global presence. By leveraging cloud services, we ensured that our system remained accessible and operational regardless of geographic location.

Furthermore, the cloud provider's infrastructure allowed us to scale our resources dynamically. This means we could allocate more computing power, storage, or networking resources during periods of high demand and scale down during quieter periods. This elasticity not only optimized system performance but also reduced operational costs, making it a cost-effective choice.

**Containerization and Orchestration:**

To streamline deployment processes and maintain consistency across different environments, we embraced containerization technology through Docker. By encapsulating our application and its dependencies into containers, we ensured that it could run consistently across various platforms and environments, from development to production.

To manage and orchestrate these containers effectively, Kubernetes was introduced. Kubernetes simplified the deployment, scaling, and management of containerized applications. It allowed us to automate tasks like load balancing, rolling updates, and self-healing, reducing manual intervention and ensuring the system's resilience and stability.

**Continuous Integration and Continuous Deployment (CI/CD):**

Incorporating CI/CD pipelines played a pivotal role in our development and deployment setup. We integrated popular CI/CD tools like Jenkins or Travis CI into our workflow. This automation pipeline ensured that code changes were automatically built, tested, and deployed to the target environment, following a well-defined workflow. This approach not only saved time but also enhanced the quality and consistency of our releases.

In conclusion, the development and deployment setup was meticulously designed to provide a solid foundation for our project. By adopting industry-standard development tools, leveraging the flexibility and scalability of cloud infrastructure, embracing containerization and orchestration, and implementing CI/CD pipelines, we created an environment that was conducive to efficient development, rapid deployment, and robust system performance. This setup laid the groundwork for the successful realization of our system for discovering and identifying available spaces.

**Development Environment:**

Our development environment was meticulously configured to promote collaboration, code quality, and efficient development practices. In addition to powerful IDEs, we also employed code review tools such as GitHub Actions and GitLab CI/CD pipelines. These automated processes ensured that every code change went through a thorough review, testing, and validation process before being integrated into the main codebase. This approach not only enhanced code quality but also reduced the chances of introducing bugs or regressions.

For version control, we followed Git best practices, including branching strategies and commit conventions. This allowed us to manage feature development, bug fixes, and releases seamlessly. We also used Git hooks to automate checks for code formatting, linting, and test coverage, ensuring that code standards were consistently maintained across the project.

**Cloud Provider Selection:**

The choice of a cloud provider was a strategic decision aimed at achieving the highest levels of reliability and scalability. We assessed various cloud providers based on factors such as data center locations, service offerings, and reputation. Ultimately, we selected a provider with a global network of data centers, enabling us to deploy our system in proximity to our target users, reducing latency, and ensuring high availability.

Additionally, the cloud provider's ecosystem offered a wide array of services, including managed databases, content delivery networks (CDNs), and machine learning resources. This rich set of services allowed us to leverage advanced functionalities without having to

**Containerization and Orchestration:**

Containerization with Docker played a pivotal role in ensuring consistent application behavior across different environments. Containers encapsulated the application along with its dependencies, ensuring that it ran the same way in development, testing, and production environments. This consistency was crucial for eliminating the "it works on my machine" problem and simplifying troubleshooting and debugging.

Kubernetes, our choice for container orchestration, provided powerful tools for managing containerized applications at scale. We defined Kubernetes manifests that specified how our application should run, including resource requirements, scaling rules, and networking configurations. Kubernetes then handled the deployment and scaling of containers automatically, ensuring that our system remained responsive under varying loads.

**Cost Optimization:**

An important consideration in our cloud deployment was cost optimization. With careful resource planning and monitoring, we optimized our cloud spending. We implemented cost management tools and practices to track resource usage, identify inefficiencies, and make informed decisions about resource allocation. This allowed us to strike a balance between performance and cost-effectiveness, ultimately benefiting our project's sustainability.

**Security and Compliance:**

Security was a top priority in our development and deployment setup. We followed industry best practices for securing cloud resources, including role-based access control (RBAC), encryption at rest and in transit, and regular security assessments. Compliance with data protection regulations, such as GDPR and HIPAA, was also a key consideration, and we ensured that our deployment setup aligned with these requirements.

In summary, our development and deployment setup encompassed a comprehensive approach to ensure the success of our project. We prioritized collaboration, code quality, and automation in our development environment. The selection of a reliable cloud provider, coupled with containerization and orchestration, ensured scalability and consistency. Cost optimization and security measures were integral components, safeguarding both our project's budget and the data it handled. This well-thought-out setup laid the foundation for a robust and efficient system for discovering and identifying available spaces.

## 5.2 Algorithms

The success of our system for discovering and identifying available spaces hinges on the utilization of advanced algorithms capable of processing large volumes of data and providing accurate results in real-time. Below, we delve deeper into the specific algorithms employed in our project:

**Computer Vision Algorithms:**

Computer vision forms the backbone of our system, allowing us to analyze images and video feeds from cameras placed in the spaces under scrutiny. Among the computer vision algorithms, we implemented:

YOLO (You Only Look Once): YOLO is a real-time object detection algorithm known for its speed and accuracy. It can identify and locate objects within an image or video frame with remarkable precision. By utilizing YOLO, our system could instantly detect the presence of vehicles, people, and other objects within the monitored spaces.

OpenCV (Open Source Computer Vision Library): OpenCV provided a versatile set of tools and functions for image and video processing. We harnessed its capabilities for tasks such as object tracking, image enhancement, and feature extraction. OpenCV's robustness and extensive feature set proved invaluable in processing visual data effectively.

**Machine Learning and Deep Learning Models:**

To enhance the intelligence of our system, we integrated machine learning and deep learning models to classify and predict data patterns. These models included:

Support Vector Machines (SVM): SVMs are powerful machine learning algorithms used for classification and regression tasks. In our system, SVMs were employed to classify objects detected by computer vision as either vehicles, pedestrians, or other relevant categories.

Random Forests: Random Forests, an ensemble learning method, were used for decision-making tasks, particularly in scenarios where multiple factors influenced space availability. These models excelled in handling complex, non-linear relationships between input data and output predictions.

Convolutional Neural Networks (CNNs): CNNs were pivotal in image analysis. We implemented deep learning models using CNN architectures to recognize patterns and anomalies in real-time visual data. This enabled us to detect events such as vehicle collisions, unauthorized access, or suspicious behavior.

Recurrent Neural Networks (RNNs): RNNs were employed for sequential data analysis, particularly in tracking the movement of objects within spaces. These models allowed us to predict future positions and behaviors, ensuring accurate identification of available spaces over time.

**Geospatial Algorithms:**

For spaces with geographical coordinates and boundaries, we leveraged geospatial algorithms to determine availability. These included:

Geographic Information Systems (GIS): GIS technology enabled us to store, analyze, and visualize geospatial data. By integrating GIS tools, we could assess space availability based on location, ensuring accurate geospatial awareness, especially in outdoor environments.

Geofencing: Geofencing techniques were utilized to create virtual boundaries and trigger notifications or actions when objects or individuals entered or exited predefined areas. This was particularly beneficial for managing access control and space allocation in secure environments.

**Data Mining and Clustering:**

In scenarios where historical data was available, we applied data mining and clustering algorithms for pattern recognition and trend analysis. These algorithms included:

K-means Clustering: K-means clustering allowed us to group similar data points together, aiding in identifying clusters of frequently available spaces or high-traffic areas.

Hierarchical Clustering: Hierarchical clustering facilitated a hierarchical view of data relationships, enabling a deeper understanding of space utilization patterns over time.

By incorporating these advanced algorithms into our system, we achieved a high degree of accuracy, real-time processing capabilities, and adaptability to various environments. These algorithms formed the core intelligence behind our solution, empowering it to discover and identify available spaces with precision and efficiency.

**Computer Vision Algorithms**:

Computer vision algorithms play a pivotal role in our system, allowing us to process visual data from cameras and sensors in real-time. The YOLO (You Only Look Once) algorithm is particularly noteworthy due to its unique ability to simultaneously detect and locate multiple objects within an image or video frame. YOLO divides the image into a grid and assigns bounding boxes to detected objects, providing precise coordinates and class labels. This real-time object detection is essential for identifying vehicles, pedestrians, or other relevant objects within the monitored spaces, enabling us to assess their occupancy accurately.

OpenCV, on the other hand, provides an extensive library of functions for image and video processing. Its utility extends to object tracking, where it can follow the movement of objects across frames, ensuring continuous monitoring of space utilization. Additionally, OpenCV assists in image enhancement, improving visibility in challenging lighting conditions or adverse weather, ultimately enhancing the accuracy of our system.

**Machine Learning and Deep Learning Models:**

Our machine learning models, including Support Vector Machines (SVM) and Random Forests, excel in classification tasks. SVMs employ a mathematical approach to determine the optimal hyperplane that separates data points into different classes. In our context, SVMs classify objects detected by computer vision, such as vehicles and pedestrians, into their respective categories, facilitating accurate space occupancy assessment.

Random Forests, as ensemble learning methods, combine multiple decision trees to make informed predictions. They are particularly advantageous when dealing with complex, multifaceted scenarios where multiple variables influence space availability. By leveraging Random Forests, our system can consider various factors simultaneously, ensuring a more comprehensive assessment of available spaces.

Convolutional Neural Networks (CNNs) are the backbone of image analysis within our system. CNNs excel in learning hierarchical features from images, enabling them to recognize intricate patterns and anomalies. We employed deep learning models based on CNN architectures to identify specific events or situations, such as vehicle collisions or unauthorized access. Their ability to learn from extensive datasets ensures a high degree of accuracy in event detection.

Recurrent Neural Networks (RNNs), known for their sequential data analysis capabilities, are invaluable for tracking object movement within spaces over time. By processing historical data, RNNs predict future positions and behaviors, facilitating the accurate

**Geospatial Algorithms:**

Geospatial algorithms, including Geographic Information Systems (GIS) and geofencing, are particularly beneficial for spaces with geographical coordinates. GIS technology provides a holistic view of spatial data, enabling us to store, analyze, and visualize geographic information. This is essential for assessing space availability based on geographical location, ensuring precise geospatial awareness, especially in outdoor environments.

Geofencing techniques establish virtual boundaries around specific areas, triggering notifications or actions when objects or individuals cross these boundaries. This functionality is instrumental in access control and space allocation, allowing us to maintain security and manage space utilization effectively.

**Data Mining and Clustering:**

Data mining techniques, such as K-means and hierarchical clustering, enhance our system's ability to identify space utilization patterns. K-means clustering groups similar data points together, making it possible to identify clusters of frequently available spaces or high-traffic areas. This insight is valuable for optimizing space allocation and resource management.

Hierarchical clustering takes clustering analysis a step further by revealing hierarchical relationships within the data. This aids in a deeper understanding of how space utilization patterns evolve over time, assisting in long-term planning and resource allocation.In summary, the advanced algorithms integrated into our system contribute to its accuracy, adaptability, and real-time processing capabilities. These algorithms leverage the power of computer vision, machine learning, deep learning, geospatial analysis, and

### 5.3 Testing

Comprehensive testing is a crucial phase in the development of our system for discovering and identifying available spaces. It ensures that the system functions reliably, meets its specified requirements, and delivers accurate results. In this section, we delve into the various testing methodologies and scenarios employed to validate the system's performance, functionality, and usability.

**Unit Testing:**

Unit testing constitutes the foundation of our testing strategy. It involves the examination of individual components and functions of the system in isolation to verify their correctness and reliability. We developed an extensive suite of unit tests using popular testing frameworks such as pytest and JUnit. These tests assessed the behavior of functions, methods, and classes across various input scenarios, ensuring that each component performed as expected.

Furthermore, we employed test-driven development (TDD) principles, where unit tests were created before writing code. This approach guided the development process, promoting code quality and reducing the likelihood of introducing defects.

**Integration Testing:**

Integration testing was instrumental in assessing the interactions and collaborations between different modules and components of the system. This testing phase ensured that data flowed smoothly between components and that the system's various parts functioned harmoniously together.

Tools like Postman and Selenium were used to conduct integration tests. Postman facilitated the testing of APIs, ensuring that endpoints provided the expected responses and that data transfers occurred without errors. Selenium, on the other hand, was employed for testing web interfaces, validating that user interactions and data inputs were

**Functional Testing:**

Functional testing was imperative for validating that the system met its specified requirements and performed its intended functions accurately. To ensure comprehensive coverage, we designed a suite of test cases that covered various use cases and scenarios.

Functional tests examined critical system functionalities, such as object detection, space identification, and notification triggering. This testing phase aimed to identify any deviations from the expected behavior, ensuring that the system met user expectations and adhered to its functional requirements.

**Performance Testing:**

Performance testing was conducted to evaluate the responsiveness and scalability of the system. It was crucial to ascertain that the system could handle varying loads without degradation in performance. Several types of performance tests were executed:

Load Testing: Load testing involved subjecting the system to simulated loads, gradually increasing the number of concurrent users or data inputs. This helped us identify bottlenecks and assess how the system handled increased traffic.

Stress Testing: Stress testing pushed the system to its limits, examining its behavior under extreme conditions, such as excessive concurrent user connections or data volume. This allowed us to identify failure points and weaknesses in the system's architecture.

Benchmarking: Benchmarking was performed to establish performance baselines and compare the system's performance against industry standards. It provided valuable nsights into areas that required optimization and fine-tuning.

**User Acceptance Testing:**

User acceptance testing (UAT) was a critical step in ensuring that the system aligned with user expectations and provided a satisfactory user experience. End-users and stakeholders actively participated in UAT, interacting with the system and providing feedback on usability, user interface design, and overall functionality.

User feedback gathered during UAT played a pivotal role in refining the system's user interface, improving accessibility, and addressing any usability concerns. It helped bridge the gap between technical implementation and real-world user needs.

In conclusion, our comprehensive testing strategy encompassed unit testing, integration testing, functional testing, performance testing, and user acceptance testing. These testing methodologies collectively ensured the reliability, accuracy, and usability of our system for discovering and identifying available spaces, ultimately leading to a robust and user-friendly solution.

# CHAPTER 6

# RESULTS AND DISCUSSION

## 6.1 System Performance Evaluation

In this section, we will present a comprehensive evaluation of the system's performance. This includes quantitative measures such as accuracy, precision, recall, and F1-score for object detection and space identification algorithms. We will also discuss the response times for real-time processing and the system's ability to handle varying loads. Any notable anomalies or challenges encountered during testing will be highlighted.

## 6.2 Space Utilization Insights

In this subsection, we will delve into the insights gained from the system regarding space utilization. We will discuss trends, patterns, and anomalies observed in the data, emphasizing how these findings can be leveraged for optimizing space allocation and resource management. Real-world examples and use cases will be provided to illustrate the practical implications of the system's results.

## 6.3 User Feedback and Usability

User feedback is a vital aspect of system evaluation. We will present the feedback obtained during user acceptance testing (UAT) and discuss how it influenced the system's user interface and overall usability. Insights from end-users and stakeholders will be used to highlight areas of improvement and future development possibilities.

### 6.4 Scalability and Resource Efficiency

Scalability is a key consideration in any system's performance evaluation. We will assess how well the system handles varying loads and its resource efficiency in terms of CPU and memory utilization. Scalability limitations and opportunities for optimizing resource allocation will be discussed.

### 6.5 Challenges and Limitations

No system is without its challenges and limitations. In this section, we will candidly discuss any challenges encountered during the development and testing phases. Limitations related to data quality, hardware constraints, or algorithmic constraints will be addressed, along with potential strategies for overcoming them in future iterations.

### 6.6 Future Enhancements and Recommendations

Building upon the findings and insights from the system's results, we will outline potential areas for future enhancements and recommendations for improving the system. This will include suggestions for algorithm refinements, user interface enhancements, scalability improvements, and integration possibilities with other technologies or systems.

### 6.7 Conclusion

In the conclusion of this chapter, we will summarize the key results, insights, and implications discussed throughout the chapter. We will also emphasize how the system's performance aligns with the project's objectives and goals, providing a clear link between the outcomes and the initial project objectives.

Chapter 6, "Results and Discussion," will serve as a critical section of your report, showcasing the achievements, challenges, and potential for further development in your

# CHAPTER 7
# CONCLUSION

## 7.1 CONCLUSION

In conclusion, the system for discovering and identifying available spaces is a valuable tool that greatly simplifies the process of finding and securing spaces for various purposes. With its advanced search capabilities and comprehensive database, it eliminates the need for time-consuming manual searches and provides users with a wide range of options tailored to their specific requirements. The system's ability to streamline communication between space providers and seekers enhances efficiency and facilitates seamless transactions. Overall, this system offers a convenient and efficient solution to the obstacles often faced when searching for available spaces, making it an indispensable resource for businesses, organizations, and individuals in need of suitable locations for their activities.

## 7.2 FUTURE WORK

In the future, there is a growing need for systems that can efficiently discover and identify available spaces in various settings, such as parking lots, offices, and public places. These systems aim to alleviate the frustration and time wasted by individuals in searching for suitable spaces. The future work in this area focuses on developing intelligent algorithms and technologies that enable real-time detection and communication of available spaces. Such systems will utilize advanced sensing techniques, including image and video processing, as well as machine learning algorithms to automatically identify vacant spaces. In addition, the integration of these systems with mobile applications and smart devices will enable users to easily access real-time information about available spaces and reserve them if needed. The future work also includes improving the accuracy and reliability of space detection and identification algorithms, considering different environmental factors and challenges. Moreover, the development of centralized platforms that can integrate multiple space discovery systems will enable a seamless experience for users in various locations and settings. Overall, future work in discovering and identifying available spaces will greatly enhance convenience, efficiency, and overall

### 7.3 Research Issues

#### 1. Data Privacy and Security:

The issue of data privacy and security has gained significant prominence in recent years, particularly in systems that involve the collection and processing of sensitive information. In the context of our system for discovering and identifying available spaces, it's essential to explore the research issues related to safeguarding privacy and ensuring data security.

One research challenge revolves around ensuring compliance with data protection regulations such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA), depending on the application domain. Balancing the need for real-time data processing with data anonymization and user consent mechanisms is an ongoing research issue.

Additionally, addressing the ethical implications of monitoring and recording individuals' movements and behaviors in public and private spaces is a subject of ongoing research. Striking a balance between enhancing security and respecting privacy rights necessitates innovative approaches and technologies, such as edge computing and federated learning, which allow data processing to occur closer to the source and without compromising privacy.

#### 2. Robustness in Diverse Environments:

The robustness of our system across diverse environments is a critical research issue. While the system may perform well in controlled conditions, real-world scenarios often present challenges such as varying lighting conditions, inclement weather, and crowded spaces.

Research in this area may involve the development of adaptive algorithms capable of adjusting to changing environmental factors. Exploring techniques like transfer learning, where models trained in one environment can adapt to another, holds promise in addressing this challenge. Additionally, the use of multimodal sensor fusion, which combines data from cameras, lidar, radar, and other sensors, can enhance robustness by providing redundant information and reducing the impact of environmental fluctuations.

**3. Real-time Processing Optimization:**

Achieving optimal real-time processing remains a crucial research issue, especially in applications where milliseconds can make a significant difference, such as autonomous vehicles or security systems. To tackle this challenge, research efforts can focus on hardware acceleration using GPUs, TPUs, or custom ASICs, which can speed up neural network inference and image processing.

Additionally, advancements in edge computing, where processing occurs closer to the data source, can reduce latency. Research in optimizing edge devices for efficient real-time analysis is an evolving area. Furthermore, exploring parallel processing techniques and distributed computing architectures can enhance the system's real-time capabilities and responsiveness.

**4. Integration with IoT and Smart Spaces:**

Integrating our system with IoT and smart space technologies presents exciting research opportunities and challenges. Research efforts can delve into creating standardized protocols and interfaces that enable seamless communication between IoT devices and our system. This includes developing interoperability standards to ensure compatibility with various IoT ecosystems.

Moreover, investigating ways to harness IoT sensor data for more precise space utilization insights is an ongoing research avenue. Smart spaces rely on sensor data to adapt to user needs and optimize resource allocation. Collaborative research between IoT and space utilization experts can lead to innovations in creating more connected and intelligent environments.

### 7.4 Implementation Issues

**1. Hardware Compatibility and Infrastructure Costs:**

Selecting compatible hardware components and managing infrastructure costs are practical challenges in system implementation. Research in this area can explore methods for optimizing hardware choices based on specific application requirements. Leveraging cloud resources, such as serverless computing, serverless databases, and serverless storage, can help mitigate hardware compatibility issues and provide cost-effective solutions.

Additionally, investigating cost estimation models and strategies for optimizing cloud resource utilization is valuable. Implementing auto-scaling and resource allocation algorithms can further enhance efficiency while reducing infrastructure costs.

**2. Algorithm Optimization and Efficiency:**

Algorithm optimization and efficiency are paramount during implementation. Ongoing research in this domain should focus on developing algorithms that not only deliver accurate results but also minimize computational resource consumption. Techniques like model quantization, pruning, and neural architecture search can contribute to more efficient deep learning models.

Furthermore, exploring edge AI solutions that offload computational tasks to edge devices can reduce the strain on central processing units and enhance overall system efficiency. Collaborative research with hardware manufacturers to design AI-specific hardware tailored to our algorithms is a promising avenue for achieving optimal performance.

**3. Data Management and Storage:**

Effectively managing and storing large volumes of data generated by cameras and sensors is a practical implementation challenge. Research in this area can investigate data compression techniques, data lifecycle management strategies, and distributed storage solutions to ensure data availability, reliability, and cost-effectiveness.

Additionally, addressing data storage scalability and data privacy concerns through techniques like differential privacy and homomorphic encryption can be explored. Ensuring that data management practices align with evolving regulatory requirements remains a critical focus of research.

**4. User Interface and Experience Design:**

User interface (UI) and user experience (UX) design play a pivotal role in the successful adoption of the system. Research efforts should continue to explore user-centered design methodologies and conduct usability testing to refine the UI/UX. Collaborative research with human-computer interaction (HCI) experts can lead to innovations in designing intuitive interfaces that cater to diverse user needs.

Moreover, research can delve into accessibility considerations to ensure that the system is usable by individuals with disabilities. Exploring inclusive design practices and integrating assistive technologies can enhance the overall user experience.

**5. Scalability and Load Balancing:**

Scalability and load balancing issues are practical concerns during system implementation. Research in this area can investigate dynamic load balancing algorithms that efficiently distribute incoming requests across multiple servers or containers. Exploring auto-scaling policies that adjust resource allocation based on real-time demand can further optimize system scalability.

Additionally, research can focus on optimizing communication and data synchronization in distributed systems, reducing bottlenecks, and ensuring seamless scaling. Investigating serverless computing platforms and their applicability for handling sudden spikes in traffic is another avenue for enhancing scalability and resource efficienc

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# APPENDIX

## A. SOURCE CODE

dashboard.py

```
import pandas as pd
import json
import plotly.express as px
import plotly.graph_objects as go
from dash import Dash, dcc, html, Input, Output, dash_table
from dash.exceptions import PreventUpdate
import data_prep.compute_health_score as chs
import data_viz.scatterplot_data as sd
import data_prep.data_cleaning as dc
import data_viz.bar_chart as bar_chart

pd.options.mode.chained_assignment = None
df = dc.tract_to_neighborhood()

external_stylesheets = [
    {
        "href": "https://fonts.googleapis.com/css2?"
                "family=Lato:wght@400;700&display=swap",
        "rel": "stylesheet",
    },
]

app = Dash(__name__, external_stylesheets=external_stylesheets)

boundaries = sd.neighborhood_zoom()
scatter_df = sd.scatter_data()
TABLE_COLS = ["Neighborhood", "Hardship Score",
            "Mental Distress",
```

```
            "Physical Distress",
            "Diabetes",
            "High Blood Pressure",
            "Life Expectancy",
            "Health Risk Score",
            "Vacant Lots",
            "Number of Green Spaces",
            "Area of Green Spaces"]

# ---------------------------------------------------------------------------
# App layout
app.layout = html.Div([
    html.Div([
        html.P(children="□□□", className="header-emoji"),
        html.H1("Less Vacant Places, More Green Spaces", style={'text-align': 'center'}, className="header-title"),
        html.P(
            children="Analyzing the effect of green spaces on public health",
            className="header-description")],
            className="header"),
    html.Br(),
    html.Div([html.Div([html.H3("Distribution across Neighborhoods in Chicago", className="map-title"),
                    html.Br(),
                    html.H4("Select a map layer:", className="subheader-title"),
                    dcc.Dropdown(id="slct_parameter",
                            options=[
                                {"label": "Hardship Score", "value": "Hardship Score"},
                                {"label": "Health Risk Score", "value": "Health Risk Score"},
                                {"label": "Number of Vacant Lots", "value": "Vacant Lots"},
                                {"label": "Number of Green Spaces", "value": "Number of Green Spaces"},
                                {"label": "Area of Green Spaces", "value": "Area of Green
```

```
Spaces"}],
                multi=False,
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            html.Br(),

            html.Div(id='health_inputs_container',
                children = [html.H4("Select at least two health indicators to
compute the health risk score.", className="subheader-title"),
                    dcc.Checklist(
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                        options={
                            "Mental Distress" : "Mental Distress",
                            "Physical Distress": "Physical Distress",
                            "Diabetes": "Diabetes",
                            "High Blood Pressure": "High Blood Pressure",
                            "Life Expectancy": "Life Expectancy"
                            },
                        value = ["Mental Distress", "Life Expectancy"],
                        className="checklist"
                    )],
                style= {'width': '50%', 'display': 'block'}
                )],
            style= {'width': '45%', 'display': 'inline-block', 'vertical-align': 'top'}
        ),
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            multi=False,
```

```
            value="CHICAGO",
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          options={
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              "Vacant Lots": "Vacant Lots"
                },
          value = ["Parks", "Vacant Lots"],
          className="checklist"
            ),
      ],
      style= {'width': '45%', 'display': 'inline-block', 'vertical-align': 'top'}
  )
  ]
),

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      dcc.Graph(id='chicago_map_scatter', figure={},
        style={'width': '50%', 'display': 'inline-block'})
]),

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html.H4("Hardship Score - Composite score incorporating unemployment,"
      "age dependency, education, per capita income, crowded housing,"
      "and poverty into a single score", className="subheader-title"),
html.H4("Health Risk Score -  Composite score incorporating at least two of"
```

```
        "the following metrics selected:", className="subheader-title"),
    html.H5("   - Mental Distress: Mental health not good for ≥14 days during the past
30 days among adults aged ≥18 years (%)", className="checklist"),
    html.H5("   - Physical Distress: Physical health not good for ≥14 days during the
past 30 days among adults aged ≥18 years (%)", className="checklist"),
    html.H5("     - Diabetes: Diabetes among adults aged ≥18 years (%)",
className="checklist"),
    html.H5("   - High Blood Pressure: High blood pressure among adults aged ≥18
years (%)", className="checklist"),
    html.H5("    - Life Expectancy: Average life expectancy at birth (years)",
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title"),

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```

```
            style={'width': '20%', 'display': 'inline-block'}),
        dcc.Graph(id='bar_healthrisk', figure={},
            style={'width': '20%', 'display': 'inline-block'}),
        dcc.Graph(id='bar_vacantlots', figure={},
            style={'width': '20%', 'display': 'inline-block'}),
        dcc.Graph(id='bar_greenspaces', figure={},
            style={'width': '20%', 'display': 'inline-block'}),
        dcc.Graph(id='bar_areagreenspaces', figure={},
            style={'width': '20%', 'display': 'inline-block'})
    ]),

    html.Br(),

    html.Div([
        dash_table.DataTable(
            id='table',
            style_cell={
                'fontSize':14,
                'font-family': 'sans-serif',
                'padding': '5px',
                'color': 'rgb(246, 234, 223)',
                'backgroundColor': 'transparent'},
            style_header={'fontWeight': 'bold'}
            )
    ]),

    html.Br()],

    style={'margin-left':'25px',
            'margin-right':'25px',
            'margin-bottom':'30px',
            'margin-top':'30px'}
)
```

```python
# ---------------------------------------------------------------------------
# Update graph showing distribution of selected parameter across Chicago
@app.callback(
    Output(component_id='chicago_map_choro', component_property='figure'),
    [Input(component_id='slct_parameter', component_property='value'),
    Input(component_id='health_inputs_checklist', component_property='value')]
)
def update_choro(option_slctd, health_params):
    if not option_slctd:
        raise PreventUpdate
    else:
        with open('data_prep/data/Community_Areas.geojson') as fin:
            neighborhoods = json.load(fin)

        choro_df = chs.append_health_score(df, health_params)

        fig = px.choropleth_mapbox(choro_df, geojson=neighborhoods,
locations='Neighborhood',
                            featureidkey="properties.community",
                            color=option_slctd,
                            color_continuous_scale="algae",
                            mapbox_style="carto-positron",
                            zoom=9, center = {"lat": 41.81, "lon": -87.7},
                            opacity=0.5,
                            labels={'df':option_slctd}
                        )
        fig.update_layout(margin={"r":0,"t":0,"l":0,"b":0})

        return fig

# Checklist for selecting health indicators to be included in calculation
```

```
# of Health Risk Score is dependent on selection of Health Risk Score as
# a map layer
@app.callback(
    Output(component_id='health_inputs_container', component_property='style'),
    Input(component_id='slct_parameter', component_property='value'))
def show_health_inputs_checklist(parameter):
    if not parameter:
        raise PreventUpdate
    else:
        if "Health Risk Score" in parameter:
            return {'display': 'block'}
        else:
            return {'display':'none'}

# Update graph showing locations of green spaces and/or vacant lots
# for a selected neighborhood
@app.callback(
    Output(component_id='chicago_map_scatter', component_property='figure'),
    [Input(component_id='slct_locations', component_property='value'),
    Input(component_id='slct_neigh', component_property='value')]
)
def update_scatter(option_slctd, neigh_slct):
    if not neigh_slct:
        raise PreventUpdate
    else:
        if "Vacant Lots" not in option_slctd:
            data = scatter_df[scatter_df['Land Use'] == "Park"]
        elif "Parks" not in option_slctd:
            data = scatter_df[scatter_df['Land Use'] == "Vacant Lot"]
        else:
            data = scatter_df


        zoom_opt = boundaries.loc[neigh_slct]["Zoom"]
```

```
    lat_opt = boundaries.loc[neigh_slct]["Lat"]
    lon_opt = boundaries.loc[neigh_slct]["Lon"]

    with open('data_prep/data/Community_Areas.geojson') as fin:
        neighborhoods = json.load(fin)

    fig = px.scatter_mapbox(data, lat="latitude", lon="longitude", color="Land Use", hover_name="name", hover_data=["community_area_name"],
                            color_discrete_map={"Park":"green", "Vacant Lot":"gray"}, size='normalized_size')
    fig.update_layout(mapbox=go.layout.Mapbox(
                style="carto-positron",
                zoom=zoom_opt,
                center_lat = lat_opt,
                center_lon = lon_opt,
                layers=[{
                    'sourcetype': 'geojson',
                    'source': neighborhoods,
                    'type': "line",
                    'opacity': 0.5
                }]))
    fig.update_layout(margin={"r":0,"t":0,"l":0,"b":0})

    return fig

# Update dropdown list options for selecting a second neighborhood to
# exclude the first selected neighborhood
@app.callback(
    Output(component_id='slct_second_neigh', component_property='options'),
    Input(component_id='slct_neigh', component_property='value')
    )
def update_second_neigh(first_neigh):
    neighborhoods = [neigh for neigh in boundaries.index if neigh != first_neigh and
```

```
neigh != 'CHICAGO']
    return neighborhoods

# Bar chart for Hardship Score
@app.callback(
    Output(component_id='bar_hardship', component_property='figure'),
    [Input(component_id='health_inputs_checklist', component_property='value'),
    Input(component_id='slct_neigh', component_property='value'),
    Input(component_id='slct_second_neigh', component_property='value')]
)
def update_hardship_bar(health_params, first_neigh, second_neigh):
    if not second_neigh:
        raise PreventUpdate
    else:
        data = chs.append_health_score(df, health_params)
        return bar_chart.create_bar_chart(data, health_params, (first_neigh,
second_neigh), 'Hardship Score')

# Bar chart for Health Risk Score
@app.callback(
    Output(component_id='bar_healthrisk', component_property='figure'),
    [Input(component_id='health_inputs_checklist', component_property='value'),
    Input(component_id='slct_neigh', component_property='value'),
    Input(component_id='slct_second_neigh', component_property='value')]
)
def update_healthrisk_bar(health_params, first_neigh, second_neigh):
    if not second_neigh:
        raise PreventUpdate
    else:
        data = chs.append_health_score(df, health_params)
        return bar_chart.create_bar_chart(data, health_params, (first_neigh,
second_neigh), 'Health Risk Score')
```

```
# Bar chart for Vacant Lots
@app.callback(
    Output(component_id='bar_vacantlots', component_property='figure'),
    [Input(component_id='health_inputs_checklist', component_property='value'),
    Input(component_id='slct_neigh', component_property='value'),
    Input(component_id='slct_second_neigh', component_property='value')]
)
def update_vacantlots_bar(health_params, first_neigh, second_neigh):
    if not second_neigh:
        raise PreventUpdate
    else:
        data = chs.append_health_score(df, health_params)
        return bar_chart.create_bar_chart(data, health_params, (first_neigh, second_neigh), 'Vacant Lots')

# Bar chart for Green Spaces
@app.callback(
    Output(component_id='bar_greenspaces', component_property='figure'),
    [Input(component_id='health_inputs_checklist', component_property='value'),
    Input(component_id='slct_neigh', component_property='value'),
    Input(component_id='slct_second_neigh', component_property='value')]
)
def update_greenspaces_bar(health_params, first_neigh, second_neigh):
    if not second_neigh:
        raise PreventUpdate
    else:
        data = chs.append_health_score(df, health_params)
        return bar_chart.create_bar_chart(data, health_params, (first_neigh, second_neigh), 'Number of Green Spaces')

# Bar chart for Area of Green Spaces
@app.callback(
    Output(component_id='bar_areagreenspaces', component_property='figure'),
```

```
    [Input(component_id='health_inputs_checklist', component_property='value'),
    Input(component_id='slct_neigh', component_property='value'),
    Input(component_id='slct_second_neigh', component_property='value')]
)
def update_areagreenspaces_bar(health_params, first_neigh, second_neigh):
    if not second_neigh:
        raise PreventUpdate
    else:
        data = chs.append_health_score(df, health_params)
        return bar_chart.create_bar_chart(data, health_params, (first_neigh,
second_neigh), 'Area of Green Spaces')

# Table of all indicators for selected neighborhood and health indicators
@app.callback(Output(component_id='table', component_property='data'),
    [Input(component_id='health_inputs_checklist', component_property='value'),
    Input(component_id='slct_neigh', component_property='value'),
    Input(component_id='slct_second_neigh', component_property='value')]
)
def update_table(health_params, first_neigh, second_neigh):
    if not first_neigh:
        raise PreventUpdate
    else:
        data = chs.append_health_score(df, health_params)
        if first_neigh == 'CHICAGO':
            cols_to_mean = [c for c in TABLE_COLS if c != "Neighborhood"]
            data.loc['mean'] = data[cols_to_mean].mean()
            data['Neighborhood'].loc['mean'] = 'CHICAGO'
        output = chs.filter_df(data, health_params, [first_neigh, second_neigh])
        output = output.round(2)

        return output.to_dict('records')
```